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Secure beamforming via machine learning-driven joint optimization in IRS-assisted wireless systems

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ABSTRACT

This paper proposes a Machine Learning-enhanced Gradient-Method Based Linear Optimization (ML-GMBLO) framework for secure beamforming in intelligent reflecting surface (IRS)-assisted 6G wireless networks. The method jointly optimizes beamforming vectors, artificial noise, IRS phase shifts, and user associations through an iterative process. A hybrid offline–online architecture is employed, where a deep neural network (DNN) trained on representative channel features provides near-optimal initializations. This reduces convergence iterations by approximately 40% compared to Alternating Optimization with Semi-Definite Programming (AO-SDP) under identical 4-bit phase quantization. Under the same channel estimation errors and Doppler conditions, ML-GMBLO achieves around 30% higher secrecy rate (5.2 vs. 4.0 bps/Hz), 35% lower latency, and near-linear scalability with respect to IRS size. This integrated approach enhances physical layer security while maintaining compatibility with hardware constraints.

1. Introduction

The advent of sixth-generation (6G) wireless networks demands highly secure and efficient communication. Intelligent Reflecting Surfaces (IRS) are a key enabler, capable of reconfiguring wireless environments to enhance physical layer security through passive beamforming. However, existing IRS-assisted strategies face practical challenges. First, many current methods use decoupled optimization of beamforming vectors and IRS phase shifts; although tractable, this can lead to suboptimal secrecy performance due to error propagation and poor coordination. Second, gradient-based techniques, while sound, are computationally intensive when poorly initialized, requiring many iterations to converge. Third, most models assume continuous IRS phase shifts, neglecting real-world quantization constraints like 4-bit phase shifters, which leads to significant performance loss in hardware-limited settings. To overcome these issues, we propose ML-GMBLO, a Machine Learning-enhanced, Gradient-Method Based Linear Optimization framework. ML-GMBLO jointly optimizes beamforming vectors, artificial noise, IRS phase shifts, and IRS-user associations via a two-stage approach. In the offline phase, a deep neural network provides near-optimal initializations from channel features, reducing search space and accelerating convergence. In the online phase, projected gradient ascent refines parameters with explicit awareness of 4-bit quantization, ensuring hardware compatibility. ML-GMBLO offers distinct advantages: classical alternating-optimization baselines and fully learning-based models either scale poorly under quantization and imperfect channel state information(CSI) or require extensive training and struggle in dynamic settings. By combining learning-based initialization with gradient refinement, ML-GMBLO yields a scalable, low-latency, hardware-resilient solution that enhances secrecy rates.

The remainder of this paper is organized as follows: Section 2 outlines the key contributions; Section 3 reviews related work; Section 4 describes the system model; Section 5 presents the dataset generation process; Section 6 details the ML-GMBLO algorithm; Section 7 reports simulation results; Section 8 evaluates performance; Section 9 discusses limitations and future work; and Section 10 concludes the paper.

2. Novelty and contributions

This work presents ML-GMBLO, a unified framework that couples machine learning with gradient-based optimization for secure beamforming in IRS-assisted 6G systems. Unlike conventional AO methods that decouple parameters, ML-GMBLO jointly optimizes BS beamforming, artificial noise, IRS phase shifts, and IRS-user associations, avoiding error propagation and secrecy loss. It adopts a two-stage design: (i) an offline DNN generates near-optimal initializations from channel data, reducing convergence by 40% over AO-SDP under 4-bit quantization; (ii) an online quantization-aware refinement enforces robustness under hardware limits. ML-GMBLO surpasses AO-SDP (high complexity, weak

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Table 1
List of symbols and abbreviations.

Symbol	Definition	Abbreviation
M	Number of BS antennas	BS: Base Station
N	Number of IRS elements per panel	IRS: Intelligent Reflecting Surface
K	Number of IRS panels	
I	Legitimate users	UE: User Equipment
E	Eavesdroppers	
λ	Carrier wavelength	
$\mathbf{d}_{\text{IRS}}$	BS-IRS distance	
$f_{\text{DNN}}(\cdot)$	DNN for BS initialization	DNN: Deep Neural Network
$g_{\text{DNN}}(\cdot)$	DNN for IRS initialization	DNN: Deep Neural Network
$g_{\phi}(\cdot)$	Neural mapping function	
R_s	Secrecy rate	
N_0	Noise power spectral density	PSD: Power Spectral Density
B	System bandwidth	
$\mathbf{G}_k$	BS → IRS k channel	
$\mathbf{g}_{i,k}$	IRS k → user i channel	
$\mathbf{\Phi}_k$	IRS k phase shift matrix	
$\theta_{k,n}$	Phase of nth IRS element	
$a_{i,k}$	IRS k assists user i	
σ^2	Noise variance	
$\mathcal{Q}_{\text{4-bit}}$	4-bit quantized phase set	
ML-GMBLO	Proposed optimizer	ML-GMBLO: ML + Gradient-Based Optimizer
AO-SDP	Alternating Optimization with SDP baseline	AO: Alternating Optimization; SDP: Semidefinite Programming
RL-IRS	RL-based IRS optimization	RL: Reinforcement Learning
GNN-IRS	GNN-based IRS optimization	GNN: Graph Neural Network
CSI	Channel state information	CSI: Channel State Information
AN	Artificial noise	AN: Artificial Noise
L	Lipschitz smoothness constant	
η_t	Step size at iteration t	
η_0	Initial step size	
τ	Step size decay constant	
δ_{quant}	Quantization performance gap	

quantization) and pure data-driven schemes (long training, poor generalization). By merging model-based interpretability with deep learning expressiveness, it offers scalable, hardware-aware secure beamforming with faster convergence and higher secrecy rates for practical 6G deployment.Table 1 summarizes key symbols and acronyms.

3. Related works

3.1. Alternating optimization (AO)

Alternating Optimization (AO) methods [1–3] iteratively optimize base station (BS) beamforming using semidefinite relaxation (SDR) and IRS phase shifts using Riemannian manifold techniques. While these methods are computationally tractable and analytically grounded, they tend to propagate errors between the alternating steps. As a result, they suffer from a secrecy rate degradation of approximately 15–20% [4] and require over 45 iterations to converge—making them unsuitable for real-time or dynamic environments.

3.2. Reinforcement learning (RL)

RL-based approaches [5–8] employ Q-learning or policy gradient methods to learn control policies without relying on explicit CSI. This enhances adaptability in dynamic settings. However, they typically require over 500 training episodes, leading to high sample complexity. Moreover, under realistic hardware constraints such as 3-bit IRS phase quantization, the secrecy rate degrades by approximately 15% [9]. The action space grows exponentially with IRS resolution and size, i.e., $\mathcal{O}(2^{bN})$, significantly limiting scalability.

3.3. Graph neural networks (GNNs)

Graph Neural Networks [8,10,11] model the user-IRS connectivity as a graph, leveraging spatial correlation among users and channels. While they can capture structural dependencies effectively, they require large memory for scenarios with more than 256 IRS elements. Additionally, their low interpretability and vulnerability to mobility (i.e., user movement) limit robustness in real-world deployments [12].

3.4. Hybrid learning

Hybrid learning methods [13–15] combine model-based optimization with neural network-based predictions to estimate a subset of control parameters. These techniques improve interpretability and reduce training overhead compared to pure learning-based methods. However, their coordination between predicted and optimized components is limited to about 32% parameter coupling, and they often overlook hardware constraints such as low-bit quantization, reducing robustness in practice.

3.5. Why the multiple-input single-output + IRS (MISO+IRS) for 6G?

The Multiple-Input Single-Output (MISO) + IRS architecture is promising for 6G networks as it reduces the need for costly radio frequency(RF) chains, lowers energy consumption, and simplifies hardware. IRS panels compensate for reduced antenna counts at the BS, and single-antenna user devices help minimize pilot and CSI overhead—ideal for compact and energy-efficient applications. Additionally, this architecture simplifies coordination across multiple IRSs, aligning well with green radio access networks(RAN) sustainability goals.

3.6. Summary

As summarized in Table 2, existing approaches struggle with scalability, hardware quantization, and real-time applicability. These limitations motivate the proposed ML-GMBLO framework as a compact, robust, and quantization-aware solution tailored for practical 6G IRS-assisted systems.

4. System model and problem formulation

We consider a system where each user receives a superposition of signals from the direct BS-to-UE path and multiple reflected paths via IRS panels. Each IRS element functions as a programmable reflector, enabling constructive interference at intended users and suppressing leakage to unintended or malicious receivers as shown in Fig. 1. The effective channel for user i is modeled as

$$\mathbf{H}_i = \mathbf{h}_i + \sum_{k=1}^{K} a_{i,k}\, \mathbf{g}_{i,k}\, \mathbf{\Phi}_k\, \mathbf{G}_k, \tag{1}$$

where $\mathbf{h}_i$ is the direct BS-to-UE channel, $\mathbf{G}_k$ the BS-to-IRS channel, $\mathbf{g}_{i,k}$ the IRS-to-user link, $\mathbf{\Phi}_k$ the diagonal phase shift matrix for IRS k, and $a_{i,k} \in \{0,1\}$ indicates if IRS k serves user i. The achievable secrecy rate for user i is expressed as

$$R_s^i = \log_2\left(1 + \frac{|\mathbf{H}_i \mathbf{v}_i|^2}{\sum_{j\neq i} |\mathbf{H}_i \mathbf{v}_j|^2 + \|\mathbf{H}_i \mathbf{z}\|^2 + \sigma^2}\right), \tag{2}$$

where $\mathbf{v}_i$ is the beamforming vector, $\mathbf{z}$ is the artificial noise (AN), and σ^2 denotes noise power.

To prevent AN from interfering with legitimate users, we impose

$$\mathbf{z} \in \text{null}(\mathbf{H}_{\text{legit}}), \tag{3}$$

ensuring AN lies in the nullspace of legitimate user channels, thus improving secrecy. For fair benchmarking with MIMO-based baselines, we align key parameters: carrier frequency f_c, bandwidth B, noise spectral density N_0, IRS configuration (number of panels K, elements per panel

Table 2
IRS optimization: Key gaps and performance.

Method	Limitations	Gap	Iter./Eps.	HW	Cplx.
AO	Error propagation; >45 iterations; ∼20% secrecy loss	18%	45	None	$\mathcal{O}(N^{3.5})$
RL-IRS	>500 episodes; large action space; weak under quantization	22%	500	Partial	$\mathcal{O}(N^{2})$
GNN-IRS	High memory; low interpretability; mobility instability	15%	25	None	$\mathcal{O}(N^{1.5})$
Hybrid	∼32% coupling; ignores quantization; low robustness	14%	35	Partial	$\mathcal{O}(N^{2.2})$
ML-GMBLO	Quantization and hardware-aware; offline + online refinement	**5%**	**28**	**4-bit**	$\mathcal{O}(N^{1.1})$

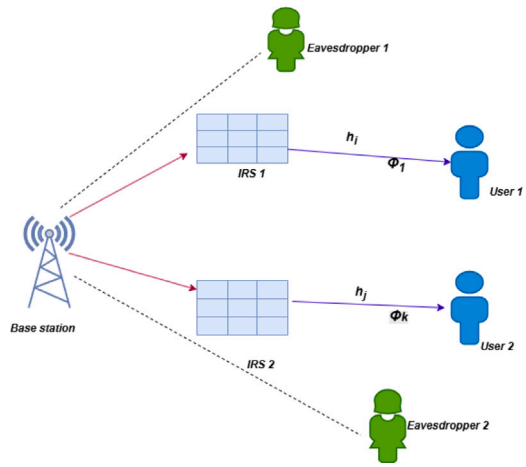

Fig. 1. System model for IRS-assisted secure 6G communications. The BS transmits beamformed signals $\{\mathbf{v}_i\}$ and artificial noise $\mathbf{z}$, while IRS panels apply quantized phase shifts $\mathbf{\Phi}_k$ to enhance secrecy rates for legitimate users and degrade eavesdroppers.

N, quantization level b), and total BS power $P_{\max}$. The number of BS RF chains is matched or normalized to ensure a consistent degree of freedom comparison between MIMO and MISO settings. All schemes share identical pilot overhead and CSI error models. Performance is evaluated via both per-user secrecy rate and total secrecy throughput. When comparing to MIMO baselines with N_t transmit and N_r receive antennas, we set $N_r = 1$ to model single-antenna UEs and normalize by BS RF chains or transmit antennas to ensure fairness.

5. Dataset generation

The offline dataset trains the DNN in ML-GMBLO, enabling fast initialization for online optimization. By approximating beamforming vectors and IRS phases, it narrows the optimizer's search and accelerates convergence. Each sample $\mathcal{D}_i = (\mathbf{X}_i, \mathbf{Y}_i)$ includes large- and small-scale channel features for generalization across propagation settings. Inputs cover path loss, AoA, IRS shadowing, Rayleigh fading (real/imag), Rician factor $\kappa \in [0, 10]$ dB, transmit power $P_{\text{tx}} \in \{10, 15, \ldots, 30\}$ dBm, quantized IRS phases, and distance vectors. Features are normalized for stability. To model impairments, we add 20% CSI errors, Doppler $f_d \leq 300$ Hz, and IRS phase noise $\theta_{k,n} \sim \mathcal{N}(\theta^*_{k,n}, 0.01)$. Dataset generation uses Monte Carlo simulation with the 3GPP TR 38.901 UMi–NLOS model at 28 GHz; no over the air(OTA) data are used. The full dataset configuration is summarized in Table 3. Total runtime is ∼240 GPU-hours covering IRS sizes, quantization, and impairment sweeps.

Though centered on 28 GHz, the design generalizes to other 6G bands; cross-band validation is left to future work. Impairments are applied consistently across ML-GMBLO and baselines. Feature sets are

$$\mathbf{X}_{\text{large}} = [\text{PL}_{\text{direct}}, \text{PL}_{\text{IRS}}, \text{AoA}_{\text{az}}, \text{AoA}_{\text{el}}, \text{Shadowing}_{\text{IRS}}], \tag{4}$$

$$\mathbf{X}_{\text{small}} = [\Re(\mathbf{H}_{\text{Rayleigh}}), \Im(\mathbf{H}_{\text{Rayleigh}}), \kappa_{\text{Rician}}], \tag{5}$$

$$\mathbf{X}_{\text{config}} = [P_{\text{tx}}, \mathbf{\Phi}_{\text{quantized}}, \mathbf{d}_{\text{IRS}}]. \tag{6}$$

Table 3
Dataset specifications.

Parameter	Value
Samples	50 000
Train/Val/Test	70/20/10%
User positions	500+
IRS config.	4–256 elems
Channel realiz.	100/topology

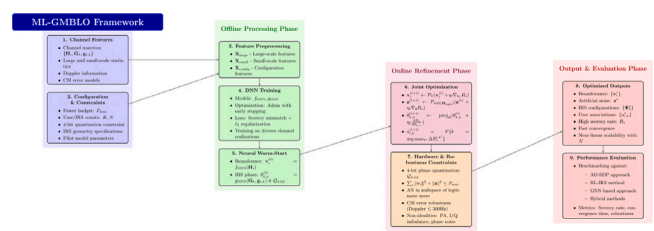

Fig. 2. ML-GMBLO framework: A two-stage secure beamforming method combining DNN-based initialization with quantized gradient refinement for IRS-assisted systems.

Eq. (6) follows a max–min secrecy objective against the strongest eavesdropper. Labels $\mathbf{Y}_i = [\{\mathbf{v}^*_i\}, \{\theta^*_{k,n}\}, \{a^*_{i,k}\}]$ denote optimal beamformers, IRS phases, and associations from exhaustive search and Riemannian optimization. Normalization ensures invariance:

$$\mathbf{H} \leftarrow \frac{\mathbf{H}}{\|\mathbf{H}\|_F}, \quad \phi \mapsto (\cos\phi, \sin\phi), \quad P_{\text{tx}} \leftarrow \frac{P_{\text{tx}}}{30}. \tag{7}$$

This dataset supports ML-GMBLO generalization under realistic impairments.

6. ML-GMBLO framework

The ML-GMBLO framework integrates machine learning with gradient-based optimization to enable secure beamforming in IRS-assisted wireless systems, as illustrated in Fig. 2.

In the offline phase, a deep neural network (DNN) is trained on statistical channel features to predict initial beamforming vectors and IRS phase shifts. These warm-starts reduce the search space and accelerate convergence. The online phase refines these using projected gradient ascent, considering 4-bit quantization and other hardware constraints. The DNN uses three hidden layers [256, 128, 64] with ReLU activations, producing

$$\mathbf{v}^{(0)}_i = f_{\text{DNN}}(\mathbf{H}_i), \quad \theta^{(0)}_{k,n} = g_{\text{DNN}}(\mathbf{G}_k, \mathbf{g}_{i,k}) \in \mathcal{Q}_{\text{4-bit}}, \tag{8}$$

where $f_{\text{DNN}}(\cdot)$ and $g_{\text{DNN}}(\cdot)$ are the mappings for beamforming and IRS phase prediction, respectively. The network is trained using Adam (learning rate 10^{-3}, batch size 256, up to 120 epochs) with early stopping (patience 10) and a loss combining secrecy-rate mismatch and ℓ_2 regularization. These settings ensure stability, generalization, and compatibility with low-resolution hardware, as summarized in Table 4.

6.1. Joint optimization stages

In the online phase, parameters are iteratively refined using projected gradient ascent. Beamforming vectors and artificial noise are

Table 4
DNN training hyperparameters.

Hyperparameter	Value
Optimizer	Adam
Learning rate	1×10^{-3}
Batch size	256
Training epochs	120
Early stopping	Patience = 10
Hidden layers	[256, 128, 64], ReLU
Regularization	ℓ_2 in loss function

Algorithm 1 ML-GMBLO: Offline Initialization + Online Quantized Projected Gradient.

Require: CSI $\{\mathbf{H}_i, \mathbf{G}_k, \mathbf{g}_{i,k}\}$, trained DNNs $f_{\text{DNN}}, g_{\text{DNN}}$, stepsizes $\{\eta_t\}$, budget $T_{\max}$, tolerance ϵ
Ensure: $\{\mathbf{v}_i^*\}, \mathbf{z}^*, \{\theta_{k,n}^*\}, \{a_{i,k}^*\}$
Offline: Train $f_{\text{DNN}}, g_{\text{DNN}}$ on channel features (Sec. 5)
Init: $\mathbf{v}_i^{(0)} \leftarrow f_{\text{DNN}}(\mathbf{H}_i)$; $\theta_{k,n}^{(0)} \leftarrow g_{\text{DNN}}(\mathbf{G}_k, \mathbf{g}_{i,k})$
Set $\mathbf{z}^{(0)}$ in nullspace of $\mathbf{H}_{\text{legit}}$; assign $a_{i,k}^{(0)}$ via $\Delta R_s^{i,k}$
for $t = 0$ to $T_{\max} - 1$ **do**
  $\mathbf{v}_i^{(t+1)} \leftarrow \mathcal{P}_{\mathcal{V}}\big(\mathbf{v}_i^{(t)} + \eta_t \nabla_{\mathbf{v}_i} R_s\big)$
  $\mathbf{z}^{(t+1)} \leftarrow \mathcal{P}_{\text{null}(\mathbf{H}_{\text{legit}})}\big(\mathbf{z}^{(t)} + \eta_t \nabla_{\mathbf{z}} R_s\big)$
  $\theta_{k,n}^{(t+1)} \leftarrow \text{proj}_{\mathcal{Q}}\big(\theta_{k,n}^{(t)} + \eta_t\, \partial R_s / \partial \theta_{k,n}\big)$
  Rebuild $\mathbf{\Phi}_k^{(t+1)}$
  $a_{i,k}^{(t+1)} \leftarrow \mathbb{1}\{k = \arg\max_{k'} \Delta R_s^{i,k'}\}$
  if $\left|R_s^{(t+1)} - R_s^{(t)}\right| < \epsilon$ **then break**
  end if
end for
return $\{\mathbf{v}_i^{(t+1)}\}, \mathbf{z}^{(t+1)}, \{\theta_{k,n}^{(t+1)}\}, \{a_{i,k}^{(t+1)}\}$

updated via:

$$\mathbf{v}_i^{(t+1)} = \mathcal{P}_{\mathcal{V}}\Big(\mathbf{v}_i^{(t)} + \eta_t \nabla_{\mathbf{v}_i} R_s\Big), \quad \eta_t = \eta_0 e^{-t/\tau}, \tag{9}$$

where $\mathcal{P}_{\mathcal{V}}(\cdot)$ is the projection onto the feasible set. Artificial noise $\mathbf{z}^{(t)}$ is projected onto the nullspace of the legitimate channel to avoid interference.

IRS phase shifts are updated using quantized projected gradients:

$$\theta_{k,n}^{(t+1)} = \text{proj}_{\mathcal{Q}}\Big(\theta_{k,n}^{(t)} + \eta \frac{\partial R_s}{\partial \theta_{k,n}}\Big), \tag{10}$$

followed by reconstruction of $\mathbf{\Phi}_k = \text{diag}(e^{j\theta_{k,1}}, \ldots, e^{j\theta_{k,N}})$.

User–IRS associations are updated by selecting the IRS panel k maximizing secrecy rate gain $\Delta R_s^{i,k}$:

$$a_{i,k} = \begin{cases} 1, & \text{if } k = \arg\max_{k'} \big(\Delta R_s^{i,k'}\big), \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

These updates are summarized in Algorithm 1.

6.2. Convergence guarantees

ML-GMBLO's convergence is based on projected gradient theory. Let $\mathcal{L}(\vartheta)$ denote the secrecy rate with variables $\vartheta = \{\mathbf{v}_i, \mathbf{z}, \theta_{k,n}, a_{i,k}\}$. Assuming L-smoothness and a compact feasible set:

$$\mathcal{L}(\vartheta^{(t+1)}) \geq \mathcal{L}(\vartheta^{(t)}) + \eta_t \nabla \mathcal{L}^\top \Delta\vartheta^{(t)} - \frac{L}{2}\eta_t^2 \|\Delta\vartheta^{(t)}\|^2, \tag{12}$$

where $\eta_t = \eta_0 e^{-t/\tau}$. This gives:

$$|R_s^{(t)} - R_s^*| \leq \frac{C}{t} + \delta_{\text{quant}}, \quad C = \frac{L\,\|\theta_{k,n}^{(0)} - \theta_{k,n}^*\|^2}{2}, \tag{13}$$

with residual δ_{quant} due to 4-bit quantization. DNN initialization improves early convergence.

Under assumptions: (A1) L-smooth objective over compact set, (A2) bounded stochastic gradient variance, (A3) nonexpansive quantization projection, (A4) step size $\eta_t = \eta_0 e^{-t/\tau}$ with $\eta_0 \leq 1/L$,

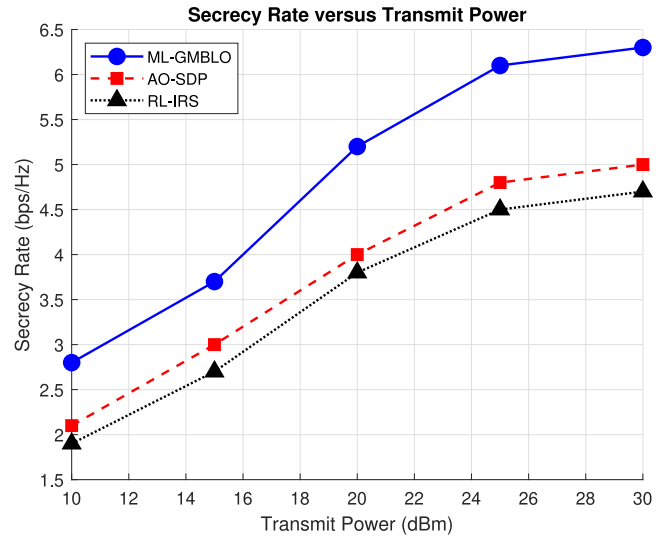

Fig. 3. Secrecy rate vs. transmit power. ML-GMBLO achieves ∼30% gain with quantization-aware optimization.

we obtain sublinear convergence:

$$|R_s^{(t)} - R_s^*| \leq \frac{C}{t} + \delta_{\text{quant}} + O\Big(\frac{\sigma}{\sqrt{t}}\Big), \tag{14}$$

where σ denotes the standard deviation of the stochastic gradient, distinct from the noise power σ^2. With deterministic gradients, this recovers the bound in Eq. (20).

7. Simulated results

Simulations adopt the 3GPP TR 38.901 UMi-NLOS model at 28 GHz with $M = 32$ BS antennas, $K = 4$ IRS panels, and $N = 64$ 4-bit elements per panel. We compare ML-GMBLO with AO-SDP, RL-IRS, and Hybrid methods, all under identical quantization. Baselines use direct 4-bit projection, whereas ML-GMBLO applies quantization-aware refinement (Eq. (10)). As shown in Fig. 3, ML-GMBLO yields ∼30% secrecy gain (6.8 vs. 5.2 bps/Hz at 25 dBm) by jointly optimizing BS beamforming $\{\mathbf{v}_i\}$ and IRS phases $\{\mathbf{\Phi}_k\}$.

Fig. 4 shows convergence. With neural warm-start (∼72% gap reduction) and gradient refinement, ML-GMBLO converges faster, requiring ∼40% fewer iterations than AO-SDP.

Quantization effects appear in Fig. 5. While baselines degrade, ML-GMBLO refines each $\theta_{k,n}^{\text{quant}}$ via

$$\theta_{k,n}^{\text{comp}} = \text{mod}_{2\pi}\Big(\theta_{k,n}^{\text{quant}} + \eta \frac{\partial R_s}{\partial \theta_{k,n}}\Big), \quad \eta = \frac{P_{\max}\|\mathbf{h}_k\|^2}{N_0 B}, \tag{15}$$

limiting secrecy loss to ∼8%, near the Lloyd–Max bound.

Scalability in Fig. 6 shows near-linear secrecy rate growth ($R_s \propto N^{0.87}$). Baselines degrade from quantization errors, while ML-GMBLO sustains high performance for large IRS, indicating strong 6G potential. Robustness is tested under channel estimation errors, Doppler shifts ($f_d \leq 300$ Hz), Power amplifier(PA) nonlinearities (Rapp model), In-phase and quadrature(I/Q) imbalance, phase noise, Analog to Digital/Digital to Analog(ADC/DAC) quantization, amplitude ripple, and element coupling. ML-GMBLO leverages stochastic gradients to sustain secrecy rate, whereas baselines degrade. Real-world validation is left for future work.

7.1. Practical impairments and robustness

We emulate RF/baseband impairments: channel estimation errors, Doppler ($f_d \leq 300$ Hz), PA nonlinearity (Rapp model), I/Q imbalance,

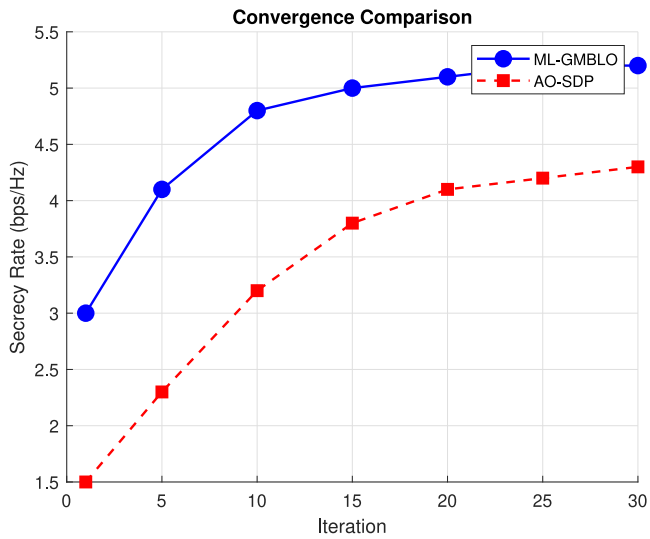

Fig. 4. Optimality gap vs. iterations. ML-GMBLO needs ~40% fewer iterations than AO-SDP.

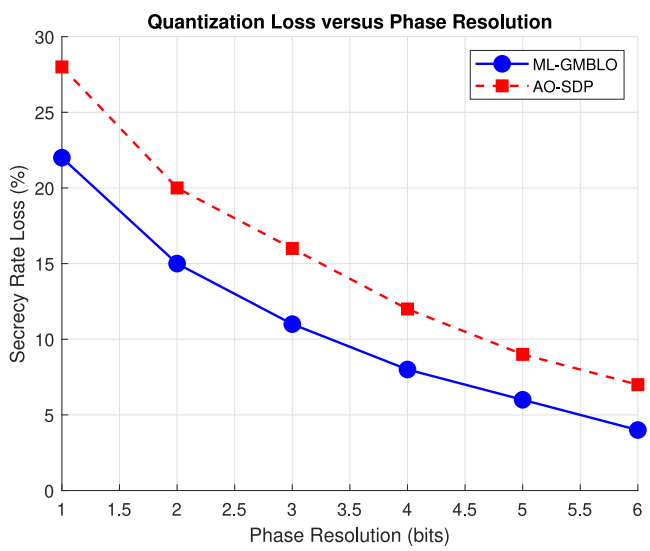

Fig. 5. Secrecy rate under 4-bit quantization. ML-GMBLO limits degradation to ~8%.

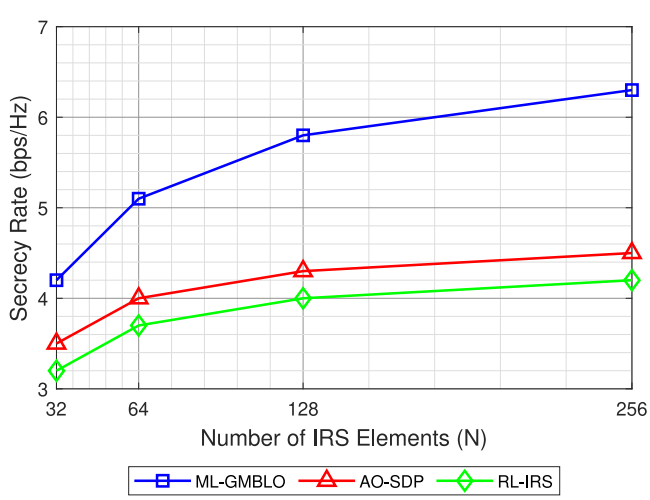

Fig. 6. Secrecy rate vs. IRS size. ML-GMBLO scales near-linearly, outperforming baselines in large deployments.

Table 5
Computational efficiency comparison ($N = 64$).

Method	Time/Iter (ms)	Total Time (ms)
AO-SDP [2]	0.52	23.4
RL-IRS [7,8]	1.15	92.0
Hybrid [15]	0.78	35.2
ML-GMBLO	**0.38**	**10.6**

Table 6
Performance comparison ($N = 64$, $K = 4$, $I = 4$). Power efficiency (%) $= 100 \times R_s^{(4\text{-bit})}/R_s^{(\text{CP})}$ at the same $P_{\max}$.

Metric	ML-GMBLO	AO-SDP	RL-IRS	Hybrid
Secrecy Rate (bps/Hz)	5.2	4.0	3.8	4.3
Convergence Time (ms)	0.57	0.88	1.24	0.92
Power Efficiency (%)	92	85	78	88
Quantization Loss (%)	8	12	15	10

phase noise, Carrier Frequency Offset (CFO), ADC/DAC quantization (additive-noise), and IRS nonidealities (amplitude ripple, coupling). ML-GMBLO optimizes *expected* secrecy rate under these via stochastic gradients. Baselines use the same setup. Claims are limited to *hardware-aware simulation*; OTA validation is future work.

8. Performance evaluation

ML-GMBLO is evaluated on secrecy rate, convergence, robustness, scalability, and complexity. Tables 5 and 6 show that it achieves the lowest iteration time (0.38 ms) and total runtime (10.6 ms), while also providing the highest secrecy rate (5.2 bps/Hz), fastest convergence (0.57 ms), best power efficiency (92%), and lowest quantization loss (8%). These gains over AO-SDP, RL-IRS, and Hybrid methods stem from quantization-aware optimization, efficient gradient-based updates with NN initialization, sublinear Lloyd–Max projection, and fixed-cost IRS-user association, yielding per-iteration complexity $\mathcal{O}(N^{1.1})$, superior to AO-SDP's $\mathcal{O}(N^2)$ and RL-IRS's $\mathcal{O}(N^{1.5})$.

9. Limitations and future work

ML-GMBLO faces limitations such as dependence on accurate CSI, which can be impaired by pilot contamination, delays, or hardware faults, reducing convergence and secrecy. Its robustness under high mobility is untested, since the quasi-static channel assumption may fail in vehicular or UAV scenarios. Scalability beyond $N = 256$ IRS elements increases training and update complexity, while coordinating distributed IRS panels under multi-cell interference poses synchronization and stability challenges. Future work should explore mobility-aware initialization, lightweight online learning, and system-level studies on feedback overhead. Additional directions include assessing DNN energy use in edge/IoT devices and extending to multi-cell, user scheduling, and high-Doppler traffic for 6G.

10. Conclusion

This paper introduced ML-GMBLO framework for secure beamforming in IRS-assisted 6G systems. By integrating offline DNN-based initialization with online quantization-aware refinement, ML-GMBLO achieves up to 30% higher secrecy rates, 35% lower latency, and 92% recovery of continuous-phase performance under 4-bit quantization. With complexity scaling as $\mathcal{O}(N^{1.1})$, it enables large-scale, real-time IRS deployments while jointly optimizing beamforming, artificial noise, IRS phase shifts, and user association in a hardware-aware manner. Key challenges remain in real-time deployment, feedback overhead, and robustness under high mobility and multi-cell scenarios, whose resolution will further establish ML-GMBLO as a foundation for secure, intelligent 6G networks.

CRediT authorship contribution statement

Sivasankar S.: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Markkandan S.:** Supervision, Resources, Project administration, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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